

ARAVIND PRADEEP

AI Engineer | Agentic Systems & RAG Pipelines | Full-Stack GenAI

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PROFESSIONAL SUMMARY

AI Engineer with two years at Perinet building the backend of an industrial-IoT platform: Python and Go services over MQTT streams, FastAPI, Docker, Kubernetes, and CI/CD, plus the model-benchmarking workstream. On my own projects I go deeper into RAG, agents, and LLMops: hybrid retrieval (BM25 + dense + cross-encoder rerank) with RAGAs and MLflow evaluation, LangGraph agent pipelines, and call-level observability. Nearly three years of professional software experience in total, including earlier enterprise work at Cognizant. Finishing my M.Sc. in AI at BTU Cottbus (thesis on Vision Transformer optimization for edge devices), available full-time immediately.

PROFESSIONAL EXPERIENCE

AI Engineer (Working Student)

Perinet GmbH | Cottbus, Germany | Jul 2024 – May 2026

- Developed Python and Go backend services connecting LLM workflows to real-time MQTT sensor streams, exposed as versioned FastAPI REST APIs with async endpoints; containerized with Docker/Kubernetes and automated CI/CD via GitHub Actions for zero-touch deployments.
- Led the containerization workstream for the AI Container umbrella project — PeriChat (Go-based chatbot with MapReduce retrieval), MessageSense (real-time MQTT message anomaly detection), and PeriXplore (sensor data exploration); profiling, QA, and deployment integration in collaboration with the engine and ML teams.
- Owned the model-benchmarking workstream; evaluated retrieval speed, generation quality, and trade-offs across model variants for the AI container platform — informed downstream chatbot and corpus design.

Software Engineer Trainee

Cognizant Technology Solutions | India | Oct 2021 – May 2022

- Developed and maintained mainframe applications (COBOL, JCL, DB2) in agile sprints; built structured debugging practices and production deployment discipline across enterprise banking systems.

PROJECTS

Multi-Agent Research & Report Pipeline | LangGraph, CrewAI, FastAPI, Docker, GitHub Actions

- Architected 3-agent pipeline (Planner, Researcher, Writer) using LangGraph state machines with strict role boundaries and Pydantic structured outputs, producing 2,000+ word research reports with verifiable source citations.
- Deployed via async FastAPI with Docker containerization and GitHub Actions CI/CD; automated end-to-end from code push to production with zero manual intervention.

RAG Evaluation System | FastAPI, Qdrant, LangChain, RAGAs, MLflow, BM25

- Engineered hybrid retrieval pipeline (dense embeddings + BM25 + Reciprocal Rank Fusion) over a 50-topic corpus using Qdrant, reaching 0.94 hit@5 and 0.96 citation presence on a 50-question evaluation set, with async embedding pre-computation and semantic caching.
- Built automated evaluation harness benchmarking faithfulness, relevance, and context recall via RAGAs; tracked 50+ prompt experiments in MLflow with automated regression alerts on retrieval quality degradation.

GraphRAG Agent | Python, networkx, LLM Entity/Relation Extraction, Docker

- Built a GraphRAG pipeline that constructs a typed, deduplicated knowledge graph from a document corpus via LLM entity/relation extraction (networkx), so retrieval follows explicit relationships instead of relying on chunk-embedding similarity alone.
- Implemented k-hop subgraph retrieval that locates query-relevant entities, expands their neighbourhood, and produces grounded answers with citations back to source, handling multi-document and relationship-spanning questions that flat RAG misses.

GraphRAG Studio | Next.js, TypeScript, React, FastAPI, Python, GraphRAG

- Built a full-stack GraphRAG application: upload documents, watch a typed knowledge graph build in real time, then chat over it with k-hop subgraph retrieval and answers cited back to source.

- Shipped a Next.js (App Router) and TypeScript front end with an interactive force-graph visualization over a FastAPI backend that wraps the graphrag-agent package.

LLM Fine-Tuning: Structured JD Extractor | QLoRA, Qwen2-0.5B, PyTorch, PEFT, 4-bit Quantization

- Fine-tuned Qwen2-0.5B with QLoRA (4-bit NF4, LoRA rank 16 / alpha 32 on attention projections) to extract structured JSON from job descriptions; reached 100% JSON validity and 70%+ entity-field accuracy on a held-out test set, trained in under 4 minutes on a 4GB GPU.
- Trained 2.16M adapter parameters (0.44% of 496M total) and evaluated per-field accuracy across entity and list fields, exposing where list-field F1 needed more training data.

LLMOps Observability Dashboard | FastAPI, React, TypeScript, PostgreSQL, Docker, GitHub Actions

- Built a full-stack LLM observability dashboard that traces every API call to PostgreSQL with latency, token counts, and per-model cost (GPT-4o, Claude, Gemini, DeepSeek) — self-hosted, no external tracing service — plus a session explorer with prompt/response previews.
- Shipped with a multi-stage Docker build, GitHub Actions CI, and 12 tests over pricing, data parsing, and API routes; React + Recharts front end reads token analytics and cost breakdowns in real time.

SKILLS

AI & Agents: LangGraph, LangChain, CrewAI, RAG, Prompt Engineering, Structured Outputs (Pydantic), Tool Use, Function Calling, MCP

LLMOps: Langfuse (Tracing, Evals), RAGAs (Faithfulness, Relevance, Recall), MLflow, LLM-as-Judge, A/B Prompt Evaluation, Guardrails

ML & NLP: Python, PyTorch, Hugging Face Transformers, scikit-learn, OpenAI/Anthropic APIs, Open-Source LLMs (Qwen, DeepSeek, Llama), Fine-Tuning, Embeddings, NER, CV

Backend & Infra: Go, FastAPI, REST/gRPC, Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Azure, AWS (S3, Bedrock, EC2), Qdrant, ChromaDB, OpenSearch, Elasticsearch, PostgreSQL, React, TypeScript

Languages: English (C1 Fluent), German (B1 Actively Improving), Malayalam (Native)

EDUCATION

M.Sc. Artificial Intelligence (Research Profile)

Brandenburg University of Technology | Cottbus, Germany | Oct 2022 – Oct 2026 (thesis phase, available full-time)

- Thesis: Content-Aware ViT Optimization on Edge Devices. Focus: ML, Computer Vision, Explainable ML, Data Mining.

B.Sc. Computer Application

BVM Holy Cross College | Kottayam, India | Jul 2018 – Mar 2021